MODULE 3 · MACHINE LEARNING · DAY 31

What Machine Learning Actually Is

Starting from zero. No maths you need to remember, no code you need to know.

5	15	0
sections	worked pictures	things to memorise

Prepared & presented by **Srinivasa Sai Chava**

Boston University

BEFORE WE START

You already use this, several times a day

Your email

sorts spam out of your inbox

YouTube

guesses the next video you want

Your phone

unlocks when it sees your face

Google Maps

predicts how long the drive takes

Your bank

flags a card payment as suspicious

Your keyboard

suggests the next word as you type

Not one of these was built by someone sitting down and writing the rules. Every one of them was LEARNED from examples. Today we find out how.

THE PLAN

Where we are going today

#	Section	The question it answers	Time
1	**Why machine learning**	Why can't we just write the rules ourselves?	12 min
2	**How a model learns**	What is actually happening when a computer "learns"?	15 min
3	**The three families**	What kinds of problems can machine learning solve?	12 min
4	**Words into numbers**	A model only does arithmetic. So what do we do with "Dell"?	18 min
5	**Put it together**	What does the whole thing look like in code?	10 min

There is no homework reading and nothing to revise first. If you can add and multiply, you have everything you need for today.

SECTION 1 OF 5

Why machine learning

Start with a job you could try to do by hand — and watch it go wrong.

- A spam filter
- Rules that break
- The flip
- The vocabulary

1.1 · THE TASK

Your job: stop spam reaching the inbox

EXAMPLE **You work at an email company. Write the rules that decide whether an email is spam.**

Your first rule	"If it contains the word FREE, it is spam."
Then you notice	"Free pizza in the lab at 5pm" is not spam. Add an exception.
Then spammers adapt	They write F-R-E-E, or FR33, or "no cost". Add more rules.

This never ends. Every trick needs a new rule, and the rules start fighting each other.

1.2 · IT BREAKS

Let's test that rule on six real emails

THE RULE: flag it as spam if it contains "free" or "won" or "urgent"

the email	rule says	truth	
WIN a FREE iPhone now!!!	SPAM	SPAM	correct
Free pizza in the lab at 5pm	SPAM	not spam	WRONG
Congratulations, you won a lottery	SPAM	SPAM	correct
Your exam results are out	not spam	not spam	correct
URGENT: verify your bank details	SPAM	SPAM	correct
Urgent: assignment due tomorrow	SPAM	not spam	WRONG

4 right out of 6. And every new trick a spammer invents needs another rule.

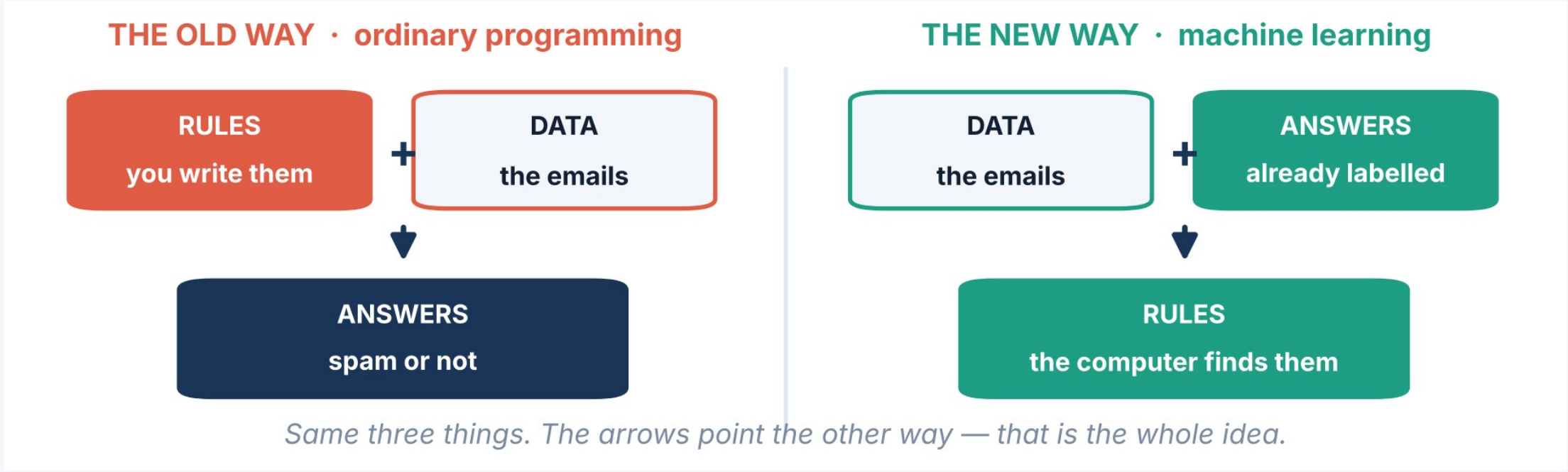
Two mistakes out of six — and both of them are honest emails sent to the spam folder. That is the worst kind of mistake you can make.

So stop writing rules.
Show the computer examples instead.

Give it 10,000 emails that people have already marked as spam or not spam, and let it work out the rules on its own.

1.3 · THE FLIP

The one picture that defines machine learning



Same three things. The arrows point the other way — that is the whole idea.

Ordinary programming: you supply the rules, the computer produces answers. Machine learning: you supply the answers, the computer produces the rules.

1.4 · VOCABULARY

The words we will use all module

these columns are what we KNOW

each one is a FEATURE

this column is what we WANT

to predict — the LABEL

one ROW = one laptop

we call it a SAMPLE

brand	RAM (GB)	screen (in)	age (yrs)	PRICE (₹k)
Dell	8	14.0	2	**34**
HP	16	15.6	1	**52**
Apple	8	13.3	3	**71**
Asus	4	14.0	4	**21**

4 samples · 4 features · 1 label. Every dataset in this course looks like this.

SAMPLE

one row

FEATURE

an input column

LABEL

the answer column

MODEL

the rule it finds

SECTION 2 OF 5

How a model learns

Nothing mysterious happens. The computer adjusts some numbers until it stops being wrong.

- A tiny example
- Shrinking the error
- Memorising vs learning
- The full pipeline

2.1 · WHAT A MODEL IS

A model is just a few numbers we can change

EXAMPLE **Five phones. We know how much RAM each has, and what it costs. Guess the price of a phone from its RAM.**

The model we will use

$$\text{price} = A + B \times \text{RAM}$$

A and B are the two numbers we get to choose.

What "training" means

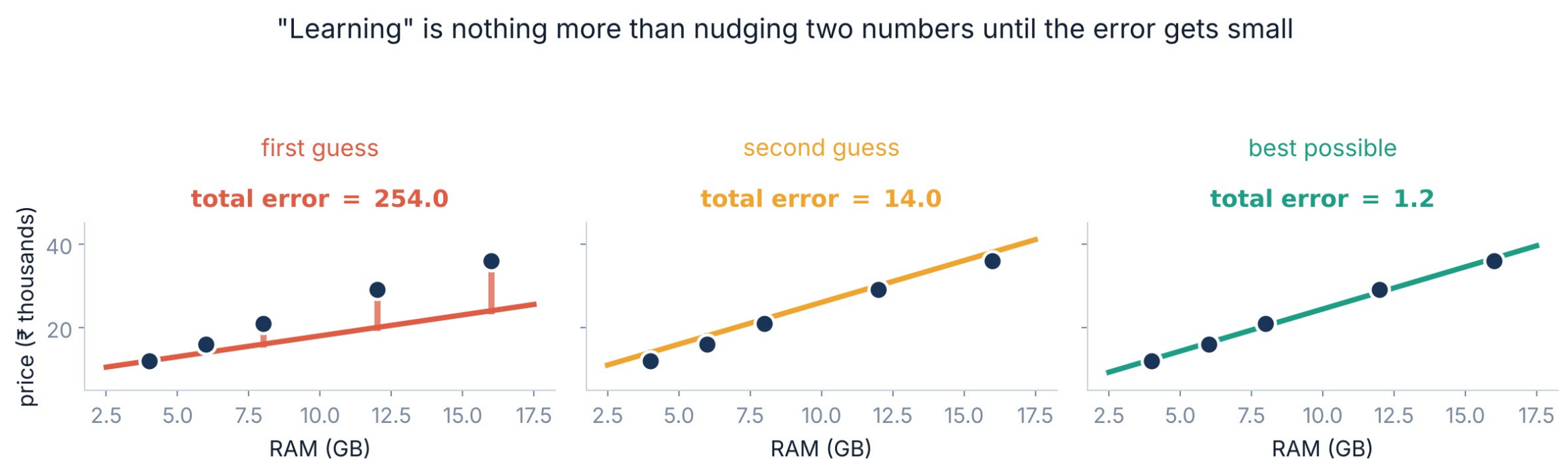
Keep changing A and B until the guesses stop being so wrong. That is all.

₹12k	→	₹16k	→	₹21k	→	₹29k	→	₹36k
4 GB of RAM		6 GB		8 GB		12 GB		16 GB

More RAM clearly costs more. But exactly how much more? That is what A and B have to capture.

2.2 · LEARNING, WATCHED

Three attempts, each less wrong than the last



The red bars are how wrong each guess was. We add up their squares to get one score.

254 → 14 → 1.2. A computer does this thousands of times a second, and it does not need to be clever about it.

That is the whole trick. Guess, measure the error, adjust, repeat.

Every model in this course does exactly this. Only the shape of the guess changes.

2.3 · THE HONEST TEST

The most important rule in machine learning

EXAMPLE **A student memorises last year's question paper and scores 100%. Did they learn anything?**

MEMORISING

Remember every answer you were shown. Perfect on the old paper. Lost on a new one.

LEARNING

Find the pattern behind the answers. Slightly imperfect on the old paper. Fine on a new one.

So we do what a good teacher does

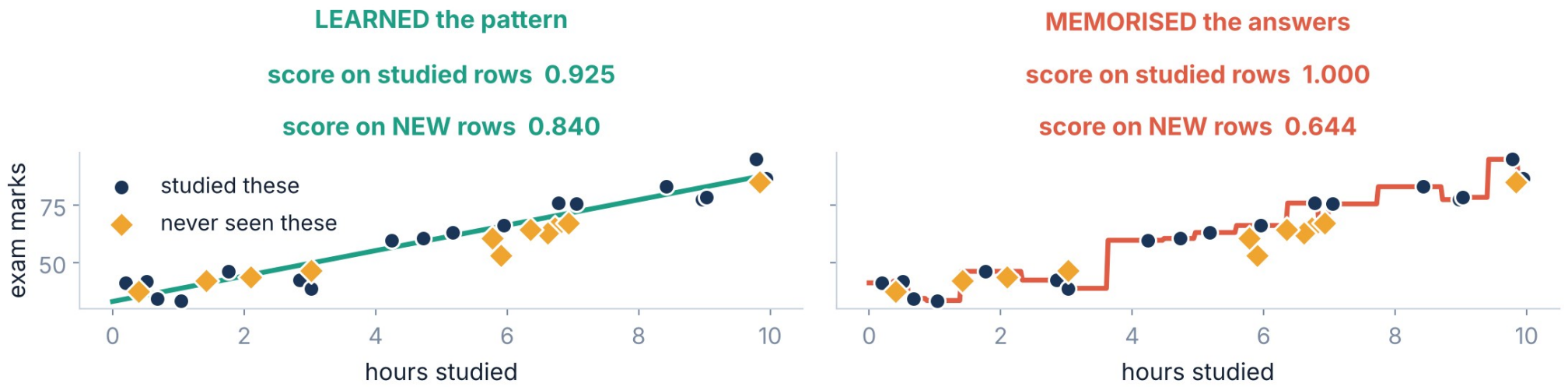
Before training, we HIDE about 30% of the rows. The model never sees them. Then we test it on exactly those rows — questions it could not possibly have memorised.

A score on rows the model has already seen is worth nothing. Ever.

2.4 · SEEING IT HAPPEN

Watch one model memorise and the other learn

A perfect score on the questions you already saw proves nothing



0.925 → 0.840

the LINE: nearly as good on new rows as on old ones

1.000 → 0.644

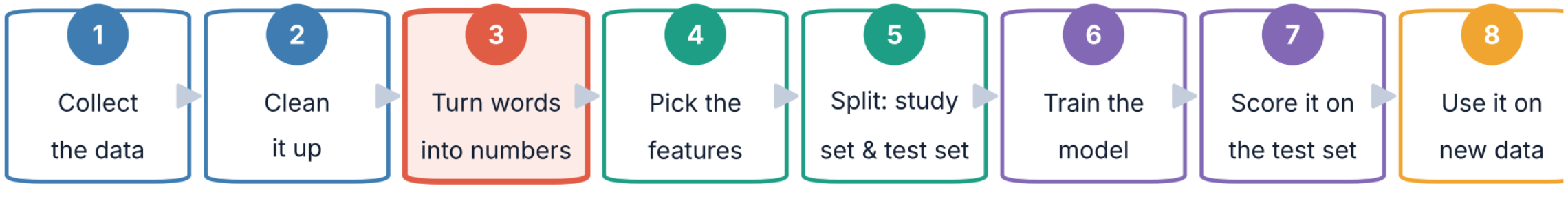
the STAIRCASE: perfect on old rows, far worse on new

0.356

how much the staircase LOST when shown new data

2.5 · THE PIPELINE

The eight steps of a machine learning project



Step 3 is today. Steps 1 and 2 are most of a real job. Step 6 is often three lines of code.

Steps 1 and 2

Collecting and cleaning the data is usually MOST of a real job.

Step 3

That is the second half of today — turning words into numbers.

Step 6

Training the model is often just three lines of code.

Nobody spends most of their time on step 6. The work is in steps 1 to 4.

SECTION 3 OF 5

The three families

Every machine learning problem you will ever meet belongs to one of three groups.

- Supervised
- Unsupervised
- Reinforcement
- Naming your problem

3.1 · THE THREE FAMILIES

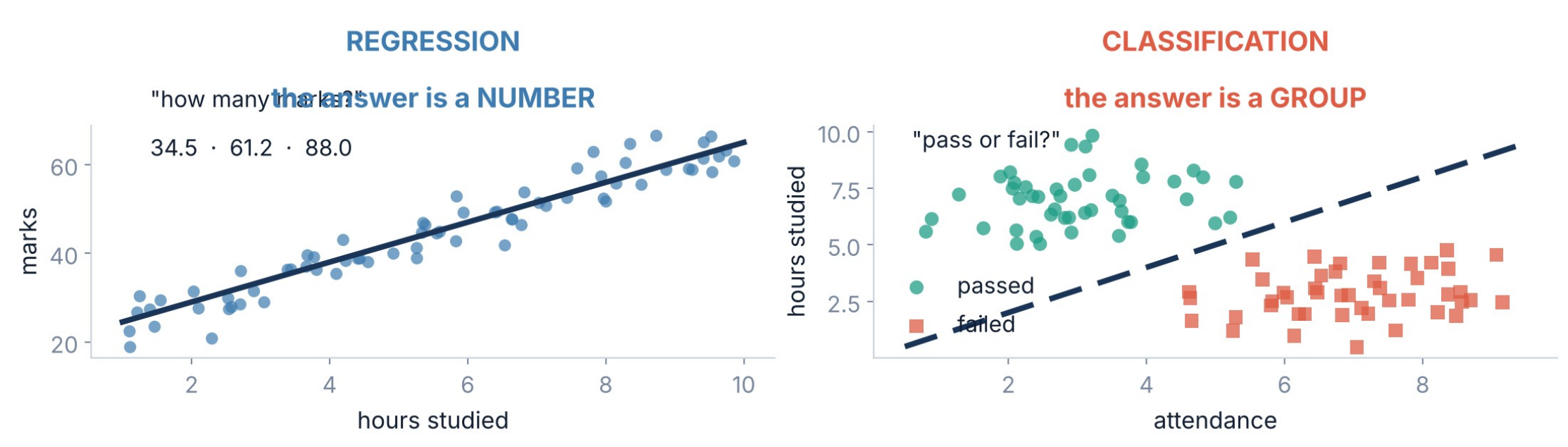
Three families, told apart by one question

SUPERVISED	UNSUPERVISED	REINFORCEMENT
We have the answers.	**We have no answers.**	**We have a score.**
1,000 emails, each already marked spam or not spam.	10,000 customers and their shopping habits. Nobody has sorted them into groups.	A game. Nobody says which move is right, but you get points when you win.
Learn the pattern, then label emails nobody has marked.	Find the groups yourself.	Learn by trying.

The question is simply: DO WE ALREADY HAVE THE ANSWERS? Yes → supervised. No → unsupervised. Only a score → reinforcement.

3.2 · NUMBER OR GROUP?

Supervised learning splits into two

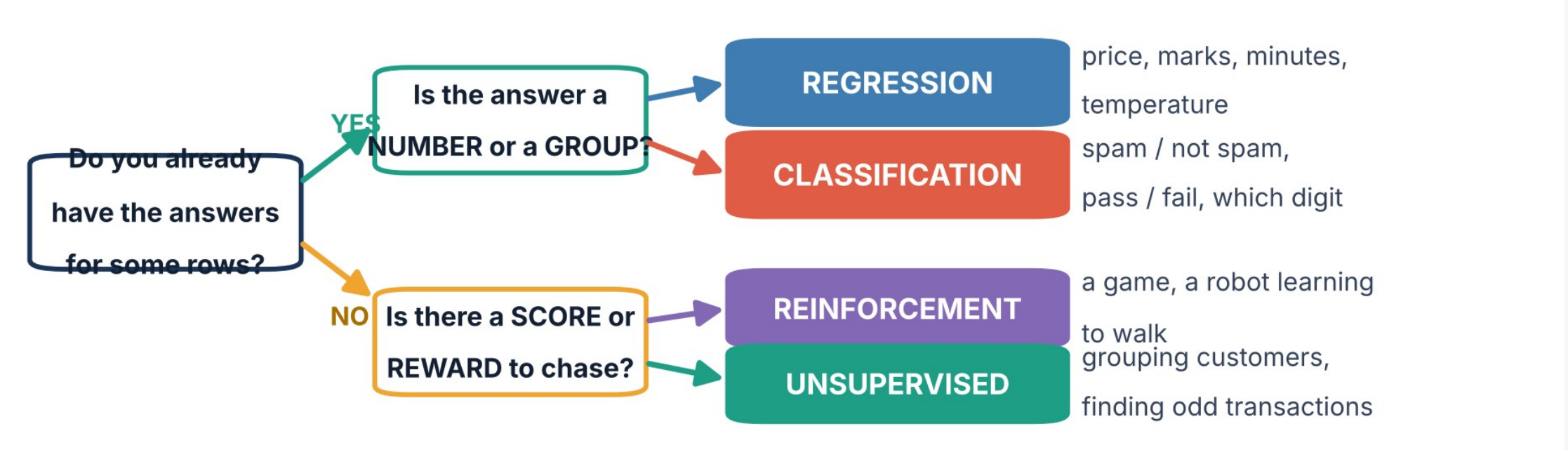


REGRESSION — the answer is a number price of a laptop · minutes to deliver · tomorrow's temperature · marks in an exam

CLASSIFICATION — the answer is a group spam or not · pass or fail · which disease · which digit is in this photo

3.3 · THE DECISION CHART

How to name any problem you meet



Two questions and you have named it. Try this on any problem anyone describes to you for the rest of the module.

3.4 · PRACTICE

Your turn — name each one

The problem	Do we have answers?	Number or group?	So it is...
Predict a flat's rent from its size	Yes, old rent records	Number	Supervised · Regression
Decide if a transaction is fraud	Yes, past cases were marked	Group	Supervised · Classification
Split 10,000 shoppers into types	No, nobody sorted them	—	Unsupervised
Teach a robot arm to pick up a cup	No, but it gets a score	—	Reinforcement
Guess how long a delivery will take	Yes, past deliveries	Number	Supervised · Regression
Read the digit in a photo	Yes, thousands are labelled	Group	Supervised · Classification

Notice how many are supervised. That is where almost all of this module lives, and it is where the rest of your course will stay.

SECTION 4 OF 5

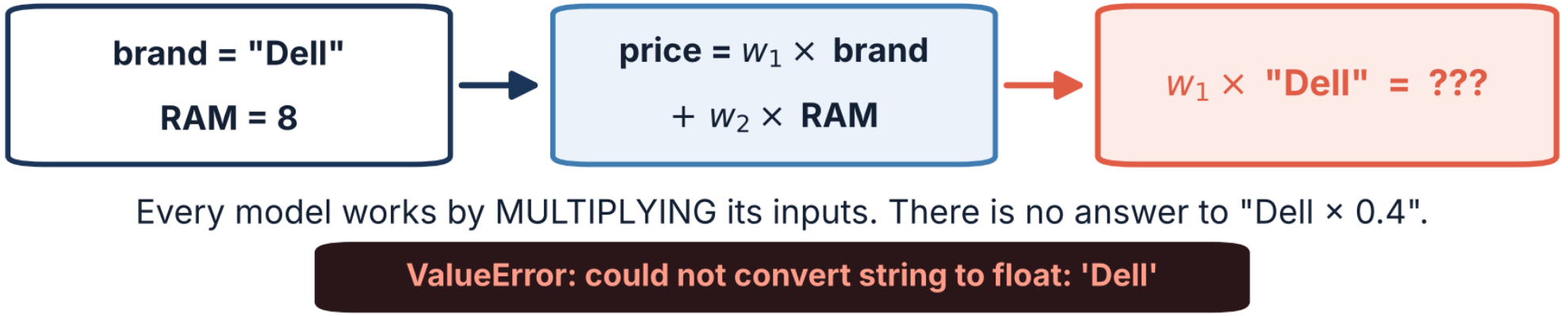
Words into numbers

A model can only add and multiply. Half your data is words. Here is how we fix that.

- Why it crashes
- Numbering: wrong
- One-hot: right
- When numbering works

4.1 · THE PROBLEM

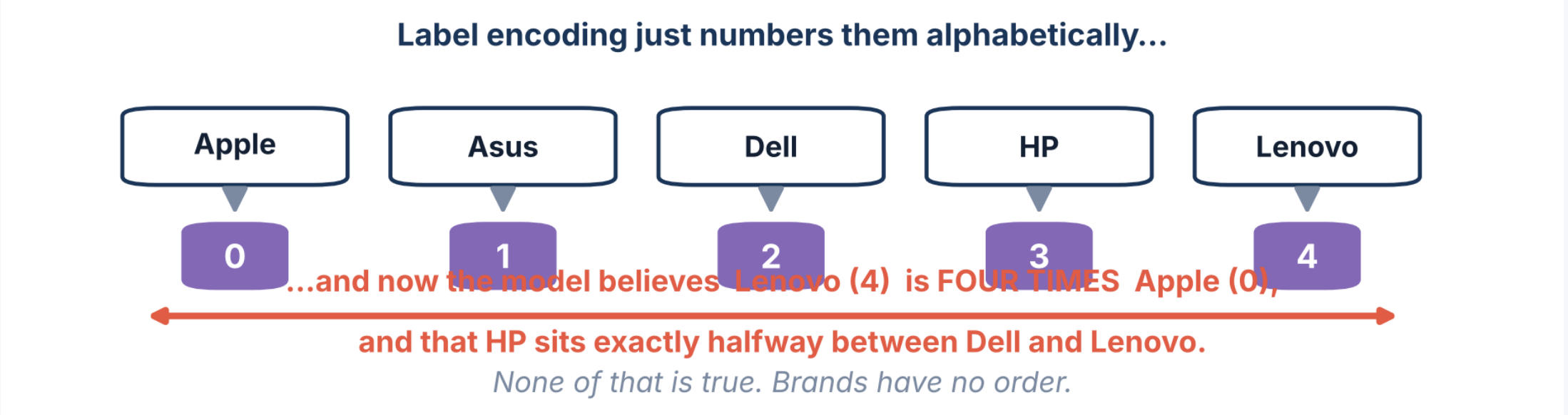
A model cannot read the word "Dell"



This is not a limitation we can argue with. Multiplication needs numbers. So every word in your data must become a number before anything else can happen.

4.2 · LABEL ENCODING

The obvious idea: just number them



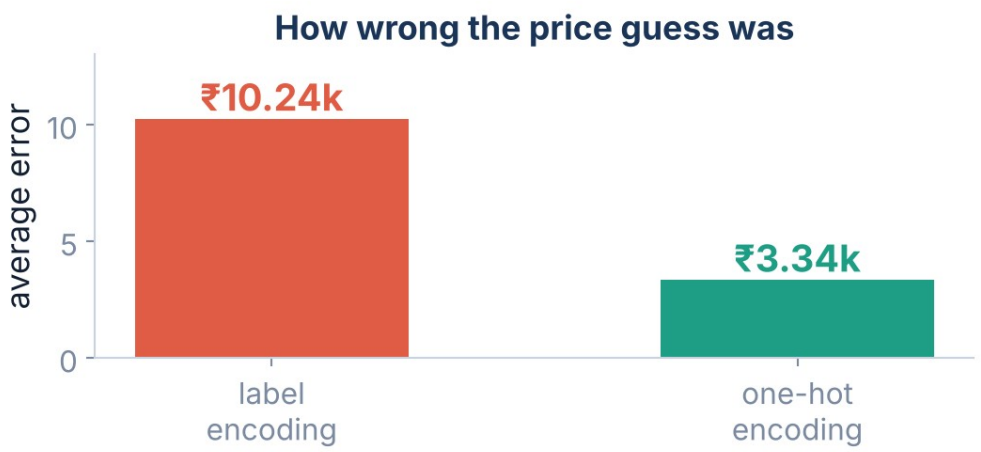
Nothing in the code warns you. The model simply believes the ladder is real and builds its whole answer on it.

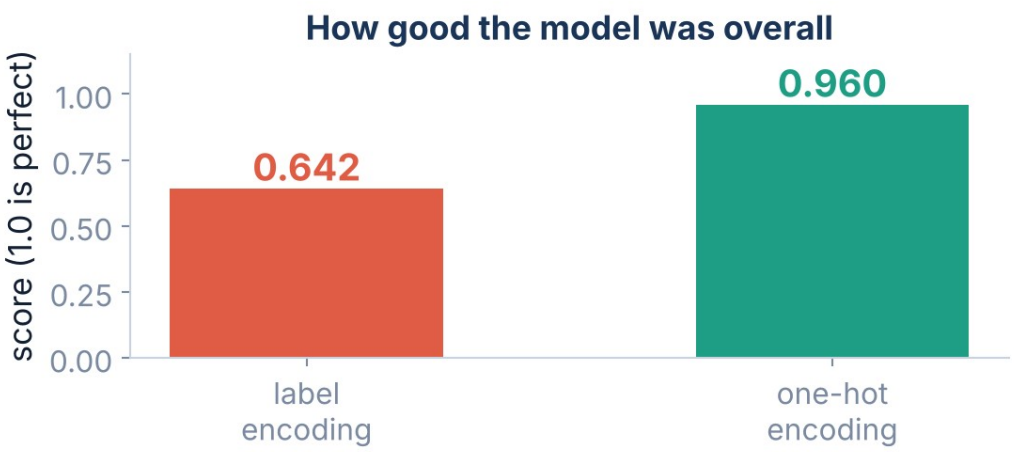
4.3 · THE DAMAGE

What that mistake costs, measured

EXAMPLE **600 laptops. Predict the price from RAM and brand. Change nothing except how the brand is encoded.**

Same data, same model, same RAM column. Only the encoding changed.





The average error TRIPLED — from ₹3,340 to ₹10,240 — purely because the brands were put in a false order.

4.4 · ONE-HOT ENCODING

The fix: one yes/no column per category

BEFORE

brand
Dell
HP
Apple
Dell

AFTER — one yes/no column per brand

is it Apple?	is it Dell?	is it HP?
0	1	0
0	0	1
1	0	0
0	1	0

Exactly one 1 in every row. No brand is bigger than any other — which is the truth.

Now no brand is bigger than another — each one simply has its own column, holding a 1 or a 0. This is called ONE-HOT encoding, and it is the default choice.

4.5 · IN PRACTICE

And it is one line of code

```
import pandas as pd

data = pd.get_dummies(data, columns=["brand"])

#  one column in   ->   three columns out
#  brand_Apple   brand_Dell   brand_HP
```

`columns=[...]`
Name every word-column you want converted. You can list several at once.

Do it to the whole table

So that train and test end up with exactly the same columns.

3 categories → 3 columns

Watch this number. It is about to become a problem.

One thing to watch

If your training data has Dell, HP and Apple, but a Lenovo turns up later, the model has no column for it and will crash. Later in the module we use a tool that handles this automatically — for now, just know it can happen.

Default rule: if the words have no natural order, use one-hot.

4.6 · WHEN ORDER IS REAL

Sometimes numbering IS the right answer

NO real order → one-hot



Is Apple "more" than Dell?

No. They are just different.

one-hot: 3 columns

REAL order → just number them



Large really IS more than Small.

numbering: 1 column, same accuracy

We measured both on 500 coffee orders: average error ₹5.25 either way.

Ask yourself one question	"Is one of these genuinely MORE than another?"
Real order → number them	Small/Medium/Large · Low/Medium/High · Class 1/2/3 · Cold/Warm/Hot

4.7 · THE TRAP

⚠️ When one-hot goes wrong



Almost every pincode appears once or twice. The model simply memorises each row.

score on rows it studied	score on NEW rows
+0.561	−0.494

With 426 columns and 900 rows, the model has almost one column per row. It stops learning anything and simply remembers who was who.

What to do instead

Keep the 10 most common pincodes and lump the rest together as "OTHER". Eleven columns instead of 426.

4.8 · THE DECISION

Which encoding should I use?

Your column looks like...	Example	Use this	Why
Words with NO order	brand, city, colour	**one-hot**	No category is bigger than another
Words WITH a real order	Small / Medium / Large	**number them 1, 2, 3**	The order carries real meaning
Only two options	yes / no, male / female	**one column of 0 and 1**	Two categories need only one column
Hundreds of options	pincode, product ID	**group the rare ones first**	Otherwise you get hundreds of columns

When you are unsure, use one-hot. It is never WRONG — it just costs extra columns.

SECTION 5 OF 5

Put it together

Everything from today, as one short program.

- The full code
- Mistakes to avoid
- Practice

5.1 · THE CODE

The whole of today, in one program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

#  STEP 3 - turn the word column into numbers
data = pd.get_dummies(laptops, columns=["brand"])

#  STEP 4 - separate the inputs from the answer
X = data.drop(columns=["price"])     # what we know
y = data["price"]                    # what we want

#  STEP 5 - hide 30% of the rows from the model
X_train, X_hidden, y_train, y_hidden = train_test_split(
        X, y, test_size=0.3, random_state=0)

#  STEP 6 - train
model = LinearRegression().fit(X_train, y_train)

#  STEP 7 - score it on the rows it never saw
print(model.score(X_hidden, y_hidden))
```

get_dummies first

Words have to become numbers before anything else works.

drop the answer

If the answer stays among the inputs, the model just copies it.

random_state=0

Makes the split identical every run, so your results repeat.

score the HIDDEN rows

This is the only number worth reporting.

5.2 · DEBUG CORNER

Six mistakes everybody makes once

#	The mistake	What you will see	The fix
1	**Feeding words straight in**	ValueError: could not convert string to float	pd.get_dummies first
2	**Numbering brands 0,1,2,3**	A model that quietly thinks Lenovo is 4× Apple	one-hot instead
3	**Scoring on studied rows**	A beautiful score that collapses in real use	Always split first
4	**Leaving the answer in X**	A suspiciously perfect score	drop the label column
5	**One-hot on 400 pincodes**	400 new columns and pure memorising	Group the rare ones
6	**Forgetting random_state**	A different score every time you run it	Set random_state=0

Numbers 1, 2 and 5 are all the same mistake: forgetting that the model only ever sees numbers, never meanings.

OVER TO YOU

Practice session — 30 minutes

A

Work it out

Name the family, read a dataset, count one-hot columns

8 questions

B

Find the mistake

Four short programs, each with one thing wrong

4 questions

C

Write the code

Encode, split, train and score end to end

5 tasks

D

Challenge

Fix a dataset nobody can train on, then catch a cheating model

2 tasks

Your notebook

`Day31_Practice_Questions.ipynb`

Part A needs no computer at all. Answer it on paper first.

Three habits to start today

- Encode words before anything else
- Hide rows before you train
- Only ever quote the hidden-row score

Part D hands you a model scoring almost perfectly and asks you to explain why it is worthless.

Day 31 — the six things to remember

The flip	Ordinary code: you give rules, it gives answers. ML: you give answers, it finds the rules.
A model is just numbers	Training means nudging those numbers until the error gets small. Nothing more.
Hide rows before training	A score on rows the model already saw proves nothing. We measured 1.000 → 0.644.
Three families	Have the answers → supervised. No answers → unsupervised. Only a score → reinforcement.
⚠️ Models cannot read words	"Dell" must become a number before anything else can happen.
⚠️ Numbering brands is wrong	It invents an order that does not exist. It tripled our error: ₹3.3k → ₹10.2k.

Next time — getting those numbers into the right SHAPE before a model sees them.

Srinivasa Sai Chava · Boston University